

- IEC 61850 MMS Protocol: Rules for Reading and Writing to IEC Addresses
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1. Overview and Scope

1.1 Purpose

This document provides a complete, authoritative reference for all rules governing the **reading and writing of IEC 61850 addresses** via the **MMS (Manufacturing Message Specification, ISO 9506)** protocol, as mapped by IEC 61850-8-1.

IEC 61850 defines an abstract communication service interface (ACSI) that is mapped to MMS for client-server communication. Understanding how ACSI object references translate to MMS variable specifications — and which operations are permitted — is fundamental to correct integration of Intelligent Electronic Devices (IEDs), SCADA systems, gateways, and engineering tools.

1.2 Scope

This document covers:

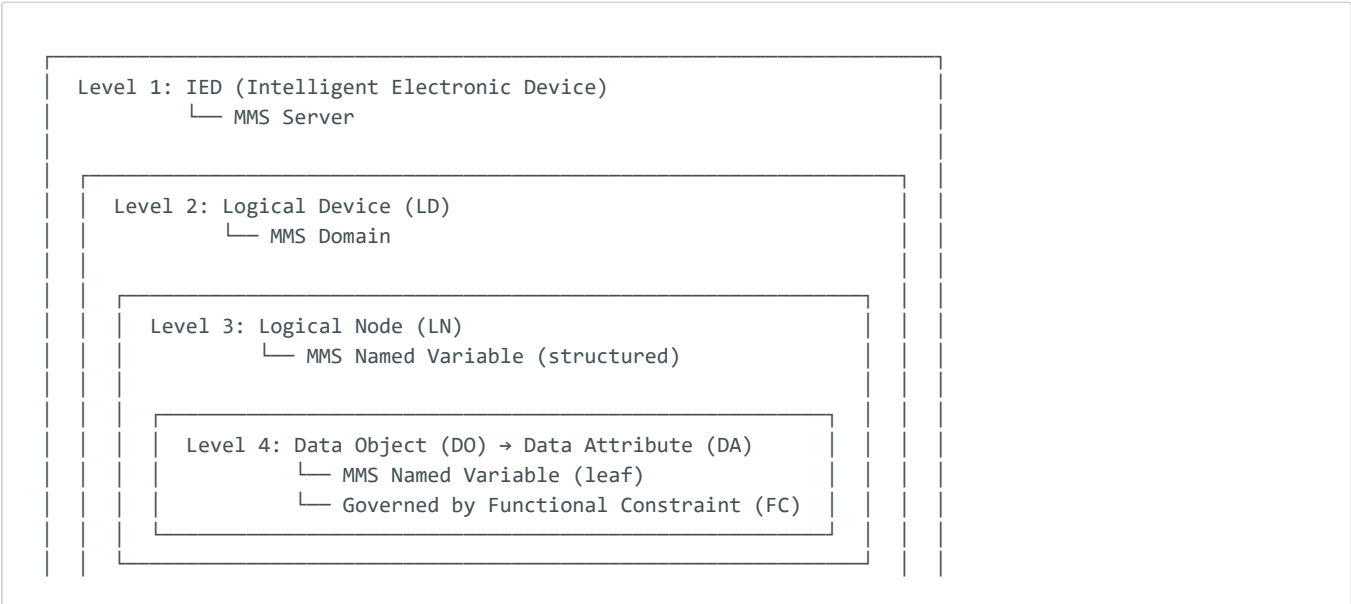
- The complete IEC 61850 information model object hierarchy
- Full address structure syntax and MMS wire encoding
- All Functional Constraint (FC) categories and their access rules
- Read rules: permitted services, reference formats, and subscription patterns
- Write rules: three distinct write mechanisms (SetDataValues, Control, Setting Group)
- Hard restrictions on non-writable attributes
- Control model state machines (Direct, SBO, Enhanced Security)
- Common Data Class (CDC) decomposition by FC
- MMS service primitive mapping
- Access control and security model
- Practical engineering examples for all scenarios

1.3 Normative References

Standard	Title
IEC 61850-7-1	Basic communication structure — Principles and models
IEC 61850-7-2	Abstract communication service interface (ACSI)
IEC 61850-7-3	Common data classes (CDC)
IEC 61850-7-4	Compatible logical node classes and data object classes
IEC 61850-8-1	MMS mapping (SCSM)
ISO 9506-1/2	Manufacturing Message Specification (MMS)

2. IEC 61850 Information Model Hierarchy

The IEC 61850 information model is a strict four-level hierarchy. Every addressable element sits at a defined level, and the level determines the MMS construct it maps to.



2.1 Level Descriptions

Level	Object	Standard Reference	MMS Construct	Example
1	IED	IEC 61850-7-1 §5	MMS Server	IED1
2	Logical Device (LD)	IEC 61850-7-1 §5.5	MMS Domain	IED1CTRL
3	Logical Node (LN)	IEC 61850-7-4	MMS Named Variable	XCBR1
4	Data Object (DO) / Data Attribute (DA)	IEC 61850-7-3, 7-4	MMS Named Variable leaf	Pos.stVal

2.2 Logical Node Classes (Selected)

Logical Node classes are defined in IEC 61850-7-4 and grouped by function domain:

LN Group	Prefix	Function	Example LNs
System	L	System-level LNs	LLN0, LPHD
Protection	P	Protection functions	PTOC, PDIF, PDIR
Protection-related	R	Auto-reclose, interlocking	RREC, RDIR
Control	C	Control functions	CSWI, CALH
Measurement	M	Analog measurements	MMXU, MMTR
Switchgear	X	Circuit breakers, disconnectors	XCBR, XSWI
Instrument Xfmr	T	Transformers	TCTR, TVTR
General	G	Generic functions	GGIO, GAPC

3. Complete Address Structure

3.1 ACSI Object Reference Syntax

The canonical IEC 61850 ACSI address has the following structure:

<code><IED_Name> / <LD_Inst> / <LN_Prefix><LN_Class><LN_Inst> . <DO_Name> . <DA_Name> [FC]</code>

Field Definitions:

Field	Description	Constraints	Example
IED_Name	IED identifier from SCL IED name= attribute	Max 64 chars, alphanumeric	IED1, PROT_A
LD_Inst	Logical Device instance from SCL LDevice inst=	Max 64 chars	CTRL, MEAS, PROT
LN_Prefix	Optional prefix to distinguish LN instances	0–11 chars	" ", MAIN
LN_Class	Logical Node class (IEC 61850-7-4)	Exactly 4 chars	XCBR, MMXU, PTOC
LN_Inst	Instance number	Numeric string	1, 2, 01
DO_Name	Data Object name (from CDC definition)	IEC 61850-7-4 defined	Pos, Hz, Op
DA_Name	Data Attribute name (may be nested)	IEC 61850-7-3 defined	stVal, mag.f, q
[FC]	Functional Constraint (implicit in context)	One of 11 valid FCs	[ST], [MX], [CO]

3.2 Address Examples — Full Decomposition

Example 1: Circuit Breaker Status

```

IED1CTRL / XCBR1 . Pos . stVal [FC=ST]
├── IED: IED1
├── LD: IED1CTRL (IED1, Control logical device)
├── LN: XCBR (Circuit Breaker), instance 1
├── DO: Position (CDC: DPC - Double Point Controllable)
└── DA: Status value (boolean CODED_ENUM)

```

Example 2: Frequency Measurement

```

IED1MEAS / MMXU1 . Hz . mag . f [FC=MX]
├── IED: IED1MEAS (Measurement logical device)
├── LN: MMXU (Measurement Unit), instance 1
├── DO: Frequency (CDC: MV - Measured Value)
├── DA structure: magnitude component
└── DA leaf: floating-point value

```

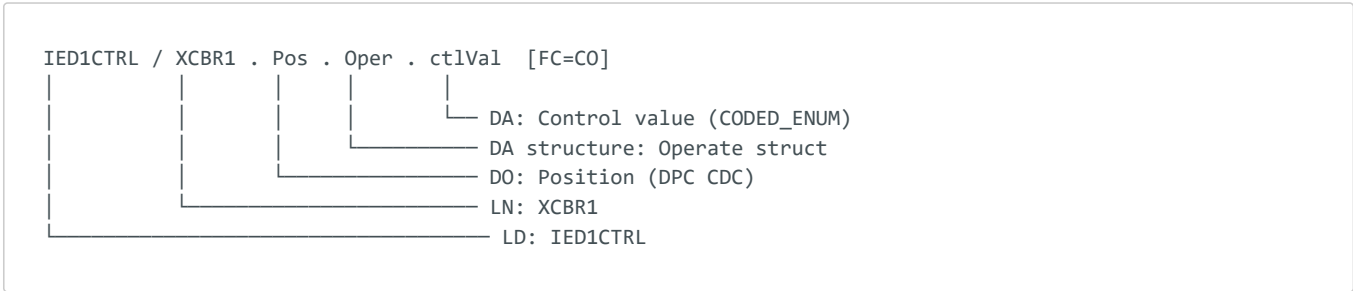
Example 3: Overcurrent Protection Setpoint

```

IED1PROT / PTOC1 . StrVal . setMag . f [FC=SP]
├── IED: IED1PROT
├── LN: PTOC (Overcurrent protection)
├── DO: Start value (CDC: ASG)
├── DA: setpoint magnitude
└── DA leaf: float setpoint value

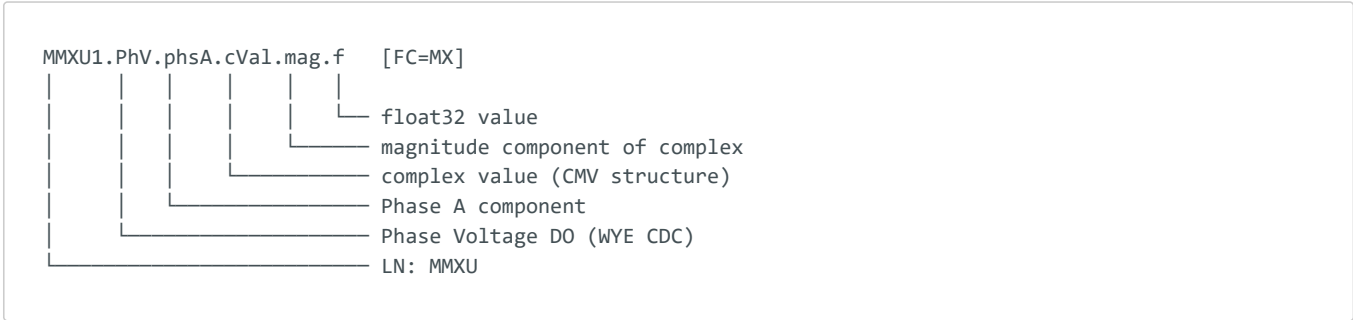
```

Example 4: Control Operation



3.3 Nested Data Attribute Paths

Data Attributes may be nested structures. The path uses . separators in ACSI notation:



3.4 Special Logical Nodes

Every Logical Device must contain two mandatory LNs:

LN	Name	Purpose	Key DOs
LLN0	Logical Node Zero	LD-level management, RCB, GoCB, SGCB	Mod, Beh, Health, NamPlt
LPHD	Physical Device	Hardware information and health	PhyNam, PhyHealth, Proxy

4. MMS Address Encoding on the Wire

4.1 ACSI-to-MMS Translation Rules

When IEC 61850 ACSI references are encoded as MMS variable specifications (IEC 61850-8-1 §12), the following rules apply:

ACSI Element	MMS Encoding	Separator
Logical Device	MMS Domain name	(domain context)
Logical Node	First segment of MMS item ID	\$
Functional Constraint	Second segment of MMS item ID	\$
Data Object	Third segment of MMS item ID	\$
Data Attribute (nested)	Subsequent segments	\$

Transformation Rule:

```
ACSI:  <LD> / <LN> . <FC_group> . <DO> . <DA>
MMS:   Domain=<IED_Name><LD_Inst>  ItemID=<LN>${<FC>}${<DO>}${<DA>}
```

4.2 Wire Encoding Examples

ACSI Reference	MMS Domain	MMS Item ID
IED1CTRL/XCBR1.Pos.stVal [ST]	IED1CTRL	XCBR1\$ST\$Pos\$stVal
IED1CTRL/XCBR1.Pos.OperctlVal [CO]	IED1CTRL	XCBR1\$CO\$Pos\$Oper\$ctlVal
IED1MEAS/MMXU1.Hz.mag.f [MX]	IED1MEAS	MMXU1\$MX\$Hz\$mag\$f
IED1PROT/PTOC1.StrVal.setMag.f [SP]	IED1PROT	PTOC1\$SP\$StrVal\$setMag\$f
IED1CTRL/LLN0.BRCB01.RptID [BR]	IED1CTRL	LLN0\$BR\$BRCB01\$RptID

4.3 MMS Variable Access Specification Structure

In MMS PDU encoding (BER/ASN.1), an IEC 61850 variable is referenced as:

```
VariableAccessSpecification ::= CHOICE {
  listOfVariable  [0] SEQUENCE OF VariableSpecification,
  variableListName [1] ObjectName
}

VariableSpecification ::= CHOICE {
  name [0] ObjectName  -- domain-specific: domain + item
}

ObjectName ::= CHOICE {
  domain-specific [1] SEQUENCE {
    domainId    Identifier,  -- e.g. "IED1CTRL"
    itemId      Identifier    -- e.g. "XCBR1$ST$Pos$stVal"
  }
}
```

4.4 Reading an Entire Data Object (All FCs)

To read all DAs of a DO across all FCs, the MMS client must issue **separate ReadRequests per FC**. There is no single MMS read that retrieves all FC partitions simultaneously.

```
Read Request 1: Domain=IED1CTRL  ItemID=XCBR1$ST$Pos      (ST partition)
Read Request 2: Domain=IED1CTRL  ItemID=XCBR1$CF$Pos      (CF partition)
Read Request 3: Domain=IED1CTRL  ItemID=XCBR1$CO$Pos      (CO partition)
```

5. Functional Constraints (FC) — The Access Control Mechanism

5.1 Principle

The **Functional Constraint (FC)** is the primary access control mechanism in IEC 61850. Every Data Attribute belongs to exactly one FC group. The FC determines:

- Whether the DA can be read
- Whether the DA can be written
- Which MMS service or ACSI service must be used
- Whether additional preconditions must be met before access

The FC is not optional metadata. It is a structural property of every DA defined in IEC 61850-7-3 CDCs and IEC 61850-7-4 DO definitions.

5.2 Complete Functional Constraint Reference

FC=ST — Status (IEC 61850-7-2 §6.2.2)

Property	Value
Full name	Status
Read	Permitted — <code>GetDataValues</code>
Write (SetDataValues)	Prohibited
Write (Control)	N/A
Source of values	Live process values updated by device firmware
Typical DAs	<code>stVal</code> , <code>q</code> (quality), <code>t</code> (timestamp)
Use case	Real-time status of primary equipment (breaker open/closed)
Notes	Values are process-driven; client writes are rejected by conformant servers

Example DAs under FC=ST:

```
XCBR1$ST$Pos$stVal    -- DPC: double-point status (intermediate/off/on/bad)
XCBR1$ST$Pos$q        -- Quality (validity, source, test, operator-blocked)
XCBR1$ST$Pos$t        -- Timestamp of last change
GGIO1$ST$Ind1$stVal    -- Boolean status input
MMXU1$ST$OpTmh$stVal  -- Operating time in hours
```

FC=MX — Measured Values (IEC 61850-7-2 §6.2.3)

Property	Value
Full name	Measured Values
Read	Permitted — <code>GetDataValues</code>
Write (SetDataValues)	Prohibited
Source of values	Transducer/ADC measurements updated by device in real-time
Typical DAs	<code>mag.f</code> , <code>ang.f</code> , <code>instMag.f</code> , <code>q</code> , <code>t</code>

Property	Value
Use case	Analog measurements: voltage, current, power, frequency
Notes	Substitution of MX values must use FC=SV, not direct write to MX

Example DAs under FC=MX:

```
MMXU1$MX$Hz$mag$f      -- Frequency magnitude (float32)
MMXU1$MX$A$phsA$cVal$mag$f  -- Phase A current magnitude
MMXU1$MX$W$phsA$cVal$mag$f  -- Active power phase A
TVTR1$MX$Vol$mag$f      -- Voltage transformer secondary output
```

FC=SP — Setpoints (IEC 61850-7-2 §6.2.4)

Property	Value
Full name	Setpoints (Set Point)
Read	Permitted — GetDataValues
Write (SetDataValues)	Permitted
MMS Service	WriteRequest
Preconditions	Client must have write access; Mod of LN must be ON
Typical DAs	setMag.f , setAng.f , setVal , maxLim , minLim
Use case	Protection thresholds, deadbands, limit values
Notes	Writes take immediate effect; no state machine required

Example DAs under FC=SP:

```
PTOC1$SP$StrVal$setMag$f  -- Overcurrent pickup level (float)
PTOC1$SP$TmMult$setMag$f  -- Time multiplier setting
MMXU1$SP$Hz$db$f          -- Frequency deadband
GGIO1$SP$AnIn1$db$f       -- Analog input deadband
```

FC=SV — Substituted Values (IEC 61850-7-2 §6.2.5)

Property	Value
Full name	Substituted Values
Read	Permitted — GetDataValues
Write (SetDataValues)	Permitted
MMS Service	WriteRequest
Preconditions	subEna must be set TRUE to activate substitution
Typical DAs	subVal , subQ , subID , subEna
Use case	Maintenance mode — substitute analog or status values for testing

Property	Value
Notes	When subEna=TRUE , the substituted value overrides the process value in reports

Example DAs under FC=SV:

```
MMXU1$SV$Hz$subVal$mag$f    -- Substitute frequency value
MMXU1$SV$Hz$subQ            -- Substitute quality flags
MMXU1$SV$Hz$subID           -- Substitutor identity string
MMXU1$SV$Hz$subEna          -- Enable substitution (BOOLEAN, write TRUE to activate)
```

Substitution Activation Sequence:

```
Step 1: Write MMXU1$SV$Hz$subVal$mag$f = 50.0    (set substitute value)
Step 2: Write MMXU1$SV$Hz$subEna = TRUE           (activate substitution)
Step 3: Reports will now show subVal with quality source=substituted
Step 4: Write MMXU1$SV$Hz$subEna = FALSE          (deactivate, restore process value)
```

FC=CF — Configuration (IEC 61850-7-2 §6.2.6)

Property	Value
Full name	Configuration
Read	Permitted — GetDataValues
Write (SetDataValues)	Permitted
MMS Service	WriteRequest
Preconditions	Client write privilege; typically requires engineering access level
Typical DAs	smpRate , smpMod , range , rangeC , units
Use case	Device configuration parameters (sampling rate, range, units)
Notes	Changes may require device restart; no ACSI state machine required

Example DAs under FC=CF:

```
TCTR1$CF$Amp$smpRate        -- Sampled value sampling rate
TCTR1$CF$Amp$smpMod          -- Sampling mode (SPS/DPS)
MMXU1$CF$Hz$range$rangeC$hhLim$f  -- High-high limit for frequency range
GGIO1$CF$AnIn1$units$SIUnit      -- SI Unit for analog input
```

FC=DC — Description (IEC 61850-7-2 §6.2.7)

Property	Value
Full name	Description
Read	Permitted — GetDataValues
Write (SetDataValues)	Prohibited

Property	Value
Source	Loaded from SCL configuration file at engineering time
Typical DAs	d (description), dU (Unicode description)
Use case	Human-readable labels for DOs and LNs
Notes	Static text set during IED configuration; not modifiable via MMS

FC=EX — Extended Definition (IEC 61850-7-2 §6.2.8)

Property	Value
Full name	Extended Definitions
Read	Permitted — GetDataValues
Write (SetDataValues)	Prohibited
Source	Vendor-defined, static
Use case	Manufacturer-specific extensions to the standard model
Notes	Definitions are fixed at manufacturing time

FC=SE — Setting Group Edit (IEC 61850-7-2 §6.2.9)

Property	Value
Full name	Setting Group Edition
Read	Permitted — GetDataValues
Write (SetDataValues)	Permitted — but only within SGCB EDIT context
MMS Service	WriteRequest (after SGCB service sequence)
Preconditions	SGCB must be in EDIT mode (SelectEditSG must be called first)
Typical DAs	Same structure as SP DAs, but stored per setting group number
Use case	Editing protection relay setting groups (up to 8 groups per LN)
Notes	Writes outside SGCB EDIT context are rejected. See Section 10.

FC=CO — Control (IEC 61850-7-2 §6.2.10)

Property	Value
Full name	Control
Read	Permitted — GetDataValues (status readback)
Write (SetDataValues)	Prohibited
Write (Control)	Permitted — via ACSI Control Model services only
MMS Services	WriteRequest to specific control DAs (\$SB0 , \$0per , \$SB0w)
Preconditions	See Section 9 for full control model state machine

Property	Value
Typical DAs	ctlVal, operTm, origin, ctlNum, T, Test, Check
Use case	Operating primary equipment: open/close breakers, raise/lower tap changers
Notes	Direct SetDataValues to CO DAs is rejected; must use control state machine

FC=OR — Operate Received (IEC 61850-7-2 §6.2.11)

Property	Value
Full name	Operate Received
Read	Permitted — GetDataValues
Write (SetDataValues)	Prohibited
Source	Internal device record, populated when an Operate is received
Typical DAs	operTm (operate timestamp), or.orIdent, or.orCat
Use case	Auditing and logging of control operations

FC=BL — Blocking (IEC 61850-7-2 §6.2.12)

Property	Value
Full name	Blocking
Read	Permitted — GetDataValues
Write (SetDataValues)	Conditional — implementation-defined
Source	Internal interlocking and blocking logic
Typical DAs	blkOpn, blkCls
Use case	Interlocking — blocking open or close operations
Notes	Write support is implementation-defined; many servers reject external writes

FC=BR / FC=RP — Buffered and Unbuffered Report Control Blocks

FC	Name	Read	Write	Purpose
BR	Buffered RCB	Yes	Yes	Configure Buffered Report Control Block on LLN0
RP	Unbuffered RCB	Yes	Yes	Configure Unbuffered Report Control Block on LLN0

Report Control Block DAs (under LLN0):

LLN0\$BR\$BRCB01\$RptID	-- Report identifier (VISIBLE_STRING, writable)
LLN0\$BR\$BRCB01\$RptEna	-- Enable reporting (BOOLEAN, writable)
LLN0\$BR\$BRCB01\$DatSet	-- Dataset reference (VISIBLE_STRING, writable)
LLN0\$BR\$BRCB01\$ConfRev	-- Configuration revision (UINT32, read-only)
LLN0\$BR\$BRCB01\$OptFlds	-- Optional fields bitmask (writable)
LLN0\$BR\$BRCB01\$BufTm	-- Buffer time in ms (writable)
LLN0\$BR\$BRCB01\$TrgOps	-- Trigger options (dchg/qchg/dupd/period) (writable)
LLN0\$BR\$BRCB01\$IntgPd	-- Integrity period in ms (writable)

LLN0\$BR\$BRCB01\$Resv -- Reservation flag (BOOLEAN, writable)
 LLN0\$BR\$BRCB01\$EntryID -- Last sent entry ID (read-only)
 LLN0\$BR\$BRCB01\$TimeOfEntry -- Timestamp of last entry (read-only)

5.3 FC Access Rights Summary Table

FC	Name	Read	SetDataValues	Control Services	SE Services	Notes
ST	Status	✓	✗	✗	✗	Process-driven, read-only
MX	Measured Values	✓	✗	✗	✗	Transducer-driven, read-only
SP	Setpoints	✓	✓	✗	✗	Direct write permitted
SV	Substituted Values	✓	✓	✗	✗	Requires subEna=TRUE to activate
CF	Configuration	✓	✓	✗	✗	Engineering access typically required
DC	Description	✓	✗	✗	✗	SCL-loaded, static
EX	Extended Definition	✓	✗	✗	✗	Vendor-static
SE	Setting Group Edit	✓	✓	✗	✓	Requires SGCB EDIT mode
CO	Control	✓	✗	✓	✗	ACSI control state machine only
OR	Operate Received	✓	✗	✗	✗	Internal record, read-only
BL	Blocking	✓	⚠	✗	✗	Implementation-defined
BR	Buffered RCB	✓	✓	✗	✗	Configure via SetDataValues
RP	Unbuffered RCB	✓	✓	✗	✗	Configure via SetDataValues
GO	GOOSE Control Block	✓	✓	✗	✗	Configure GoCB
SG	Setting Group Control	✓	⚠	✗	✓	Limited writes via SGCB services

6. Rules for Reading IEC 61850 Addresses

6.1 Fundamental Reading Rule

All Data Attributes are readable regardless of Functional Constraint, subject to access control configuration in the SCL file and server implementation.

6.2 MMS Read Service

The ACSI `GetDataValues` service maps to the MMS `ReadRequest` PDU.

MMS ReadRequest structure:

```
ReadRequest ::= SEQUENCE {
    variableAccessSpecification  VariableAccessSpecification,
    specWithResult               [0] BOOLEAN DEFAULT FALSE
}
```

Response — MMS ReadResponse:

```
ReadResponse ::= SEQUENCE {
    variableAccessSpecification [0] VariableAccessSpecification OPTIONAL,
    listOfAccessResult          SEQUENCE OF AccessResult
}
```

6.3 Granularity of Read Operations

IEC 61850 supports reading at multiple levels of granularity:

Read Scope	ACSI Service	Description	Example MMS Item ID
Single DA	<code>GetDataValues</code>	Leaf attribute only	<code>XCBR1\$ST\$Pos\$stVal</code>
All DAs of DO (one FC)	<code>GetDataValues</code>	Entire FC partition of a DO	<code>XCBR1\$ST\$Pos</code>
All FC partitions of LN	<code>GetDataObjectDirectory</code>	Enumerate all DOs	<code>XCBR1</code>
Named Variable List	<code>GetDataSetValues</code>	Pre-defined dataset	<code>IED1CTRL/LLN0\$DATASET01</code>

6.4 Quality (q) and Timestamp (t) Reading

Every measurement or status DA is accompanied by mandatory quality and timestamp DAs under the same FC:

```
Pos$stVal  [FC=ST]  -- The value
Pos$q      [FC=ST]  -- Quality descriptor (see 6.4.1)
Pos$t      [FC=ST]  -- Timestamp (UTC, with time quality)
```

6.4.1 Quality Bit Field Structure (IEC 61850-7-3 §6.5)

Bit	Name	Meaning
0-1	validity	0=good, 1=invalid, 2=reserved, 3=questionable
2	overflow	Value exceeds measurement range
3	outOfRange	Value outside defined range
4	badReference	Reference value unavailable
5	oscillatory	Value oscillating abnormally
6	failure	Equipment or communication failure
7	oldData	Value not updated in expected time
8	inconsistent	Inconsistency between redundant values
9	inaccurate	Value inaccurate
10	source	0=process, 1=substituted
11	test	Value is from test mode
12	operatorBlocked	Operator has blocked updates

6.5 Subscription-Based Reading — Report Control Blocks (RCB)

For real-time monitoring, polling via `GetDataValues` is not recommended for operational systems. The preferred mechanism is **event-driven reporting** via RCBs.

Trigger Options (TrgOps):

Trigger	Bit	Description
dchg	0	Data change — value changes
qchg	1	Quality change — quality flags change
dupd	2	Data update — any update (even if same value)
period	3	Integrity period — periodic regardless of change
gi	4	General interrogation — client-initiated snapshot

RCB Configuration Sequence:

```

1. Read LLN0$BR$BRCB01$Resv      -- Check if RCB is available
2. Write LLN0$BR$BRCB01$Resv = TRUE -- Reserve RCB
3. Write LLN0$BR$BRCB01$RptID = "CLIENT_001"
4. Write LLN0$BR$BRCB01$DatSet = "IED1CTRL/LLN0$DATASET_STATUS"
5. Write LLN0$BR$BRCB01$TrgOps = dchg|qchg|period
6. Write LLN0$BR$BRCB01$IntgPd = 60000    (60-second integrity)
7. Write LLN0$BR$BRCB01$RptEna = TRUE      (enable reporting)
8. Write LLN0$BR$BRCB01$GI = TRUE          (request general interrogation)

```

6.6 Reading Entire Logical Node Data

To read all process data from a Logical Node, issue `GetDataValues` at the LN level per FC:

```
ReadRequest: Domain=IED1CTRL ItemID=XCBR1$ST -- All ST data
ReadRequest: Domain=IED1CTRL ItemID=XCBR1$MX -- All MX data (if applicable)
ReadRequest: Domain=IED1CTRL ItemID=XCBR1$SP -- All SP data
```

6.7 Read Access Control

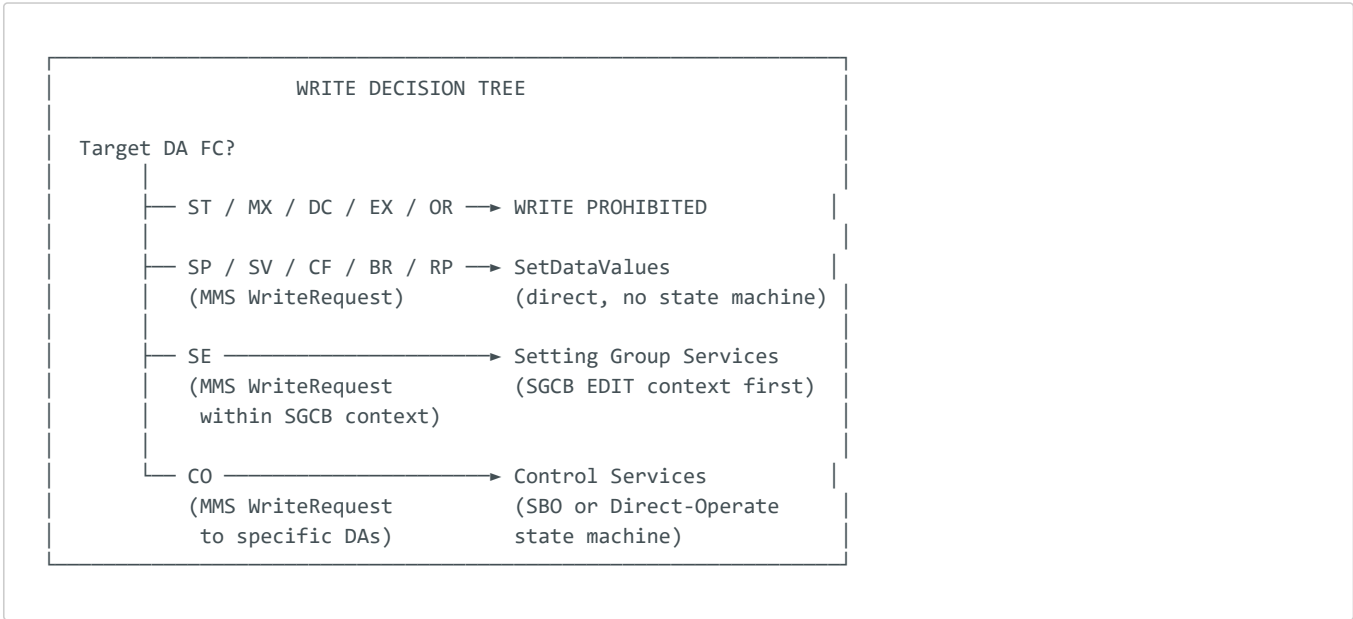
Read access may be restricted by:

- 1. **SCL AccessControl element** in `ICD/CID` files — limits which clients (IP ranges, certificate subjects) can read specific DAs
- 2. **Server implementation policy** — some vendors restrict DC and EX access to local engineering ports
- 3. **Authentication/Authorization** (IEC 62351-8) — role-based access control (RBAC) with roles: Viewer, Operator, Engineer, Installer, SecurityAdministrator

7. Rules for Writing IEC 61850 Addresses

7.1 Overview — Three Distinct Write Mechanisms

Writing to IEC 61850 addresses uses three fundamentally different mechanisms depending on the target FC:



7.2 Mechanism 1 — Direct SetDataValues (SP / SV / CF)

7.2.1 MMS WriteRequest

The ACSI `SetDataValues` service maps to the MMS `WriteRequest` PDU.

MMS WriteRequest structure:

```
WriteRequest ::= SEQUENCE {
    variableAccessSpecification VariableAccessSpecification,
    listOfData                  SEQUENCE OF Data
}
```

7.2.2 Write Preconditions

Before writing to SP/SV/CF DAs, all of the following must hold:

Condition	Check	How to Verify
LN Mode	<code>Mod.stVal</code> must be <code>on</code> (1) or <code>test</code> (3)	Read <code>XCBR1\$ST\$Mod\$stVal</code>
Write Privilege	Client must have write role	Server authentication/ACL
Data Type Match	Written data type must match DA type	Per CDC definition
Value Range	Value must be within defined <code>range</code>	Read <code>\$CF\$xxx\$range</code> first

7.2.3 MMS Write Response Handling

MMS Result	Meaning	Action
<code>success</code>	Write accepted and applied	Confirm with subsequent read
<code>object-access-denied</code>	Insufficient privileges	Check access role
<code>object-value-invalid</code>	Type mismatch or range error	Check value and type
<code>object-attribute-inconsistent</code>	Precondition not met (e.g. LN blocked)	Check <code>Mod</code> and <code>Beh</code>
<code>object-non-existent</code>	Address does not exist	Verify address in SCL

7.2.4 Direct Write Examples

Write a protection setpoint (SP):

```
Target: IED1PROT/PTOC1.StrVal.setMag.f [FC=SP]
MMS:    Domain=IED1PROT ItemID=PTOC1$SP$StrVal$setMag$f
Data:    FLOAT32 = 1.25
Expected response: success
```

Activate value substitution (SV):

```
Target: IED1MEAS/MMXU1.Hz.subVal.mag.f [FC=SV]
MMS:    Domain=IED1MEAS ItemID=MMXU1$SV$Hz$subVal$mag$f
Data:    FLOAT32 = 50.00

Target: IED1MEAS/MMXU1.Hz.subEna [FC=SV]
MMS:    Domain=IED1MEAS ItemID=MMXU1$SV$Hz$subEna
Data:    BOOLEAN = TRUE
```

8. What Cannot Be Written — Hard Restrictions

8.1 Absolutely Non-Writable FCs

The following FCs **cannot be written under any circumstances** via `SetDataValues`:

FC	Reason	What to Do Instead
ST	Reflects live process state — device firmware-maintained	Use SV substitution for testing; use CO for control actions
MX	Analog measurement from transducer — hardware-driven	Use SV substitution for testing
DC	Static text from SCL engineering tool	Modify SCL file and re-configure the IED
EX	Vendor firmware static	Contact vendor; not modifiable remotely
OR	Internal operational record of received operates	Read-only audit log

8.2 FC=CO — Cannot Use SetDataValues

Writing directly to FC=CO DAs via `SetDataValues` is **explicitly prohibited** by IEC 61850-7-2. A conformant server will respond with `object-access-denied` or `type-unsupported`.

The only correct write path is through the ACSI Control Model (see Section 9).

8.3 Attempting to Write Quality or Timestamp

Quality (`q`) and Timestamp (`t`) DAs are process-maintained attributes. Attempting to write these will be rejected:

```
Write to XCBR1$ST$Pos$q → REJECTED (object-access-denied)
Write to XCBR1$ST$Pos$t → REJECTED (object-access-denied)
Write to MMXU1$MX$Hz$q → REJECTED (object-access-denied)
```

The only mechanism to affect quality is through substitution (`FC=SV`, write `subQ`).

8.4 Summary: Write Rejection Causes

Rejection Cause	MMS Error Code	Root Cause
Non-writable FC	<code>object-access-denied</code>	Attempt to write ST, MX, DC, EX, OR
Wrong write mechanism for CO	<code>object-access-denied</code>	Direct <code>SetDataValues</code> on control DA
Client has no write role	<code>object-access-denied</code>	IEC 62351-8 RBAC — missing role

Rejection Cause	MMS Error Code	Root Cause
LN is blocked/offline	<code>object-attribute-inconsistent</code>	<code>Mod.stVal</code> is not ON
Type mismatch	<code>object-value-invalid</code>	Wrong MMS data type sent
Value out of range	<code>object-value-invalid</code>	Value exceeds <code>range</code> definition
SE write outside EDIT mode	<code>object-access-denied</code>	SGCB not in EDIT mode
Address does not exist	<code>object-non-existent</code>	Incorrect address construction

9. Control Model (FC=CO) — Detailed Write Rules

9.1 Overview

FC=CO DAs represent **control of primary equipment** (circuit breakers, disconnectors, tap changers, annunciators). The IEC 61850 control model is intentionally complex to enforce safety, authorization, and interlocking.

The `ctlModel` attribute (FC=CF) of a Data Object defines which control model applies:

ctlModel Value	Name	Description
0	<code>status-only</code>	No control possible
1	<code>direct-with-normal-security</code>	Direct operate, no confirmation sequence
2	<code>sbo-with-normal-security</code>	Select-Before-Operate required
3	<code>direct-with-enhanced-security</code>	Direct operate with additional checks
4	<code>sbo-with-enhanced-security</code>	SBO with additional checks

9.2 Control DA Structure (DPC CDC Example)

For a Double Point Controllable (DPC) Data Object such as `Pos`:

```
Pos [DO - DPC CDC]
├─ stVal    [FC=ST] -- Current process status (double-point enum)
├─ q        [FC=ST] -- Status quality
├─ t        [FC=ST] -- Status timestamp
├─ SBO      [FC=CO] -- Select (for SBO models): write to initiate select
├─ SBOw     [FC=CO] -- Select-with-value (enhanced SBO)
├─ Oper     [FC=CO] -- Operate structure
│   └─ ctlVal -- Control value (FALSE=open, TRUE=close for DPC)
│       └─ origin
│           ├── orCat    -- Origin category (0=not-supported to 6=maintenance)
│           └─ orIdent   -- Origin identifier (string)
│       └─ ctlNum       -- Control number (sequence counter)
│       └─ T            -- Timestamp of command
│       └─ Test         -- Test flag (TRUE = test, not real operation)
│           └─ Check     -- Interlock check bitmask (synchrocheck, interlockcheck)
├─ Cancel   [FC=CO] -- Cancel pending operation
└─ ctlVal   [FC=CO] -- Last commanded value (readable)
```

9.3 Direct Operate — ctlModel = 1 or 3

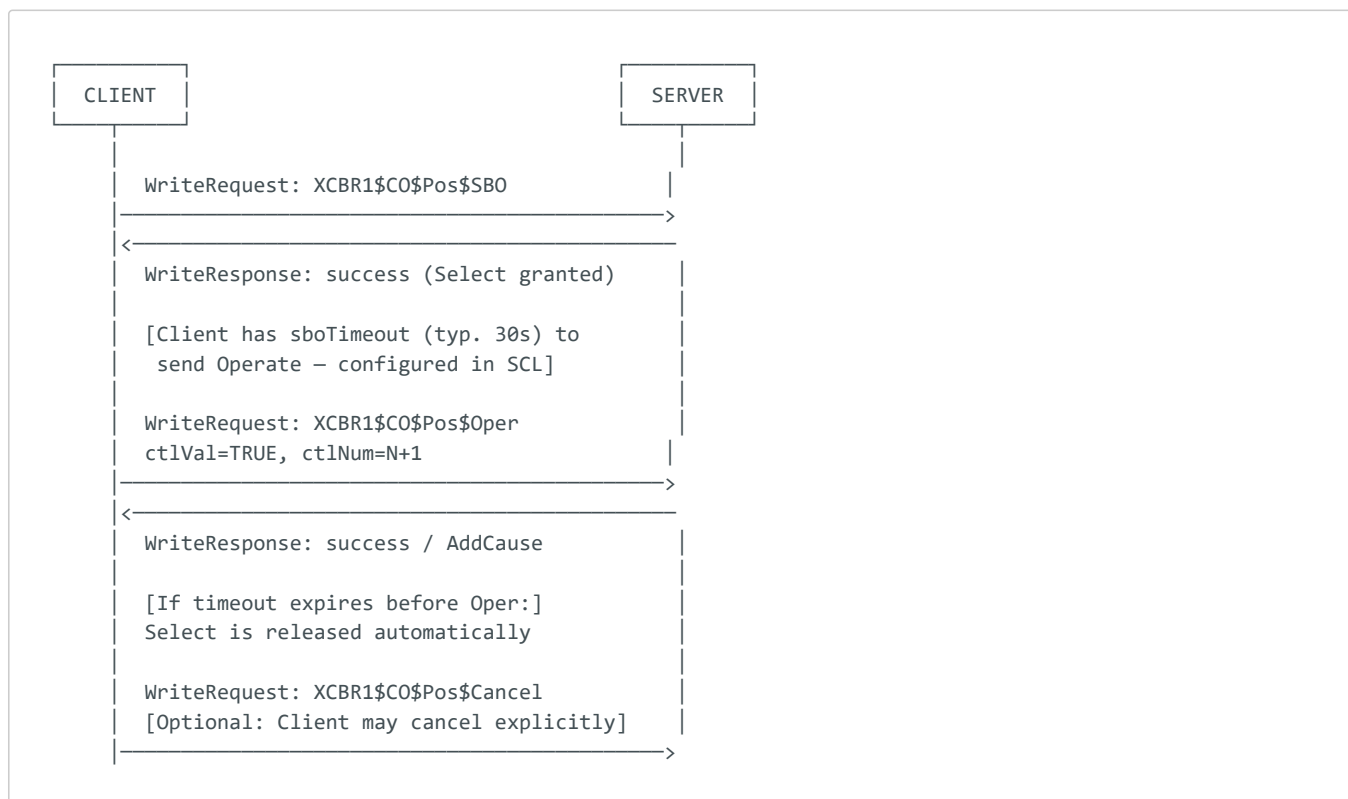


AddCause values on rejection:

AddCause	Meaning
0	Unknown
1	Not supported
2	Blocked by switching hierarchy
3	Select failed
4	Invalid position
5	Position reached
6	Parameter change in execution
7	Step limit
8	Blocked by mode
9	Blocked by process
10	Blocked by interlocking
11	Blocked by synchrocheck
12	Command already in execution
13	Blocked by health
14	1-of-n control
15	Abortion by cancel
16	Time limit over
17	Abortion by trip
18	Object not selected

AddCause	Meaning
19	Object already selected
20	Blocked by object being set

9.4 Select-Before-Operate — ctlModel = 2 or 4



9.5 Enhanced Security (ctlModel 3 and 4)

Enhanced security adds:

1. **SBOw (Select-with-Value)**: Client includes **ctlVal** in the Select, server validates the intended action
2. **Timestamp validation**: Server checks that **T** (command timestamp) is within an acceptable window (typically ± 1 second of server time — requires time synchronisation)
3. **Test mode separation**: **Test=TRUE** commands are executed in test mode only, do not affect real process

9.6 Time-Activated Operation (operTm)

For scheduled operations, **Oper.operTm** can be set to a future UTC timestamp:

```

XCBR1$C0$Pos$Oper$operTm = 2024-06-01T14:00:00.000Z
XCBR1$C0$Pos$Oper$ctlVal = TRUE
  
```

The device will execute the operation at the specified time. IED time synchronisation (SNTP/PTP per IEC 61850-9-3) is mandatory for this feature.

10. Setting Group Services (FC=SE) — Write Rules

10.1 Overview

Protection relays maintain multiple **Setting Groups** (up to 8 groups) that can be switched adaptively based on network topology. Writing to FC=SE DAs requires a dedicated service sequence through the Setting Group Control Block (SGCB) on LLN0.

10.2 SGCB Structure

LLN0\$SG\$SGCB\$NumOfSG	[FC=SG] -- Number of setting groups configured (read-only)
LLN0\$SG\$SGCB\$ActSG	[FC=SG] -- Currently active setting group (1-N)
LLN0\$SG\$SGCB\$EditSG	[FC=SG] -- Currently being edited (0=none)
LLN0\$SG\$SGCB\$CnfEdit	[FC=SG] -- Confirm edit (write TRUE to commit)
LLN0\$SG\$SGCB\$LActTm	[FC=SG] -- Last activation timestamp

10.3 Setting Group Edit Sequence

Step 1: Read	LLN0\$SG\$SGCB\$NumOfSG	-- Determine available groups (e.g. 3)
Step 2: Read	LLN0\$SG\$SGCB\$ActSG	-- Determine current active group (e.g. 1)
Step 3: Write	LLN0\$SG\$SGCB\$EditSG = 2	-- Select group 2 for editing
	→ Server enters EDIT mode for SG2	
Step 4: Write	PTOC1\$SE\$StrVal\$setMag\$f = 1.50	-- Write new pickup value to SG2
Step 5: Write	PTOC1\$SE\$TmMult\$setMag\$f = 0.50	-- Write new TMS to SG2
	[All FC=SE writes now apply to SG2]	
Step 6: Write	LLN0\$SG\$SGCB\$CnfEdit = TRUE	-- Confirm and commit edits
	→ Server exits EDIT mode; SG2 values saved	
Step 7: [Optional] Write	LLN0\$SG\$SGCB\$ActSG = 2	-- Switch active group to SG2

Critical rule: If CnfEdit is not written (TRUE) before the MMS connection closes, the edit session is discarded and SG2 retains its previous values.

11. Common Data Classes (CDC) and Their FC Decomposition

IEC 61850-7-3 defines CDCs. The following table shows the FC decomposition for the most common CDCs used in substation automation.

11.1 Status CDCs

CDC	Name	ST DAs	MX DAs	SP DAs	CO DAs	CO Structure
SPS	Single Point Status	stVal, q, t	—	—	—	—

CDC	Name	ST DAs	MX DAs	SP DAs	CO DAs	CO Structure
DPS	Double Point Status	stVal, q, t	—	—	—	—
INS	Integer Status	stVal, q, t	—	—	—	—
ENS	Enumerated Status	stVal, q, t	—	—	—	—
BCR	Binary Counter Reading	actVal, frVal, q, t	—	—	—	—
HST	Histogram	val[], q, t	—	—	—	—

11.2 Controllable CDCs

CDC	Name	ST DAs	CO DAs	ctlModel
SPC	Single Point Controllable	stVal, q, t	SB0/Oper/ctlVal	CF attribute
DPC	Double Point Controllable	stVal, q, t	SB0/SB0w/Oper/Cancel	CF attribute
INC	Integer Controllable	stVal, q, t	SB0/Oper/setVal	CF attribute
ENC	Enumerated Controllable	stVal, q, t	SB0/Oper/ctlVal	CF attribute
BSC	Binary Step Controllable	valWTr, q, t	SB0/Oper/valWTr	CF attribute
ISC	Integer Step Controllable	valWTr, q, t	SB0/Oper/valWTr	CF attribute

11.3 Measurement CDCs

CDC	Name	MX DAs	SP DAs (deadband)
MV	Measured Value	mag.f, instMag.f, q, t	db, zeroDb, range
CMV	Complex Measured Value	cVal.mag.f, cVal.ang.f, q, t	db, zeroDb
SAV	Sampled Analogue Value	instMag.f, q	—
WYE	AC Three-Phase Wye	phsA, phsB, phsC, neut, net, res	—
DEL	AC Three-Phase Delta	phsAB, phsBC, phsCA	—

11.4 Setting CDCs

CDC	Name	SP DAs	SE DAs
ASG	Analogue Setting	setMag.f, setMag.i	Same, per setting group
CURVE	Curve Shape Setting	setCrv, numPts, xUnit, yUnit	Same
CSG	Curve Shape Setting (alt.)	setCrv, numPts	Same
SPG	Single Point Setting	setVal	Same
ING	Integer Setting	setVal	Same
ENG	Enumerated Setting	setVal	Same

12. MMS Service Primitives Mapping

12.1 Complete ACSI-to-MMS Service Mapping

ACSI Service	MMS Service	PDU Type	FC Used
GetServerDirectory	GetNameList	Domain list	—
GetLogicalDeviceDirectory	GetNameList	Named variable list	—
GetLogicalNodeDirectory	GetNameList	Named variable list	—
GetDataDirectory	GetNameList	Named variable	—
GetDataDefinition	GetVariableAccessAttributes	Type description	—
GetDataValues	Read	ReadRequest/Response	All FCs
SetDataValues	Write	WriteRequest/Response	SP/SV/CF/BR/FP
GetDataSetValues	Read (named var list)	ReadRequest/Response	—
SetDataSetValues	Write (named var list)	WriteRequest/Response	—
CreateDataSet	DefineNamedVariableList	Define request	—
DeleteDataSet	DeleteNamedVariableList	Delete request	—
GetDataSetDirectory	GetNamedVariableListAttributes	Attributes request	—
Report (server→client)	InformationReport	Unsolicited info	BR/RP
GetBRCBValues	Read	ReadRequest	BR
SetBRCBValues	Write	WriteRequest	BR
GetURCBValues	Read	ReadRequest	RP
SetURCBValues	Write	WriteRequest	RP
Select	Write	WriteRequest to SBO	CO
SelectWithValue	Write	WriteRequest to SBOw	CO
Operate	Write	WriteRequest to Oper	CO
Cancel	Write	WriteRequest to Cancel	CO
CommandTermination	InformationReport	Unsolicited	CO
SelectActiveSG	Write	WriteRequest	SG
SelectEditSG	Write	WriteRequest (EditSG)	SG
SetSGValues	Write	WriteRequest	SE
ConfirmEditSGValues	Write	WriteRequest (CnfEdit)	SG
GetEditSGValues	Read	ReadRequest	SE
GetSGCBValues	Read	ReadRequest	SG

12.2 MMS Connection Establishment

Before any read or write, the MMS connection must be established with IEC 61850-specific parameters:

```
MMS Initiate Request:
  proposedMaxServOutstandingCalling: 5
  proposedMaxServOutstandingCalled: 5
  proposedDataStructureNestingLevel: 8    -- IEC 61850-8-1 requires min 8
  mmsInitRequestDetail:
    proposedVersionNumber: 1
    proposedParameterCBB: [str1, vnam, valt, vsca, tpy, vlis, real, cei]
    servicesSupportedCalling: [status, getNameList, identify, read, write,
                              getVariableAccessAttr, defineNamedVarList,
                              getNamedVarListAttr, deleteNamedVarList,
                              obtain-file, file-rename, read-journal,
                              write-journal, initialize-journal,
                              report-journal-status, create-journal,
                              delete-journal, getCapabilityList,
                              fileOpen, fileRead, fileClose,
                              fileRename, fileDelete, fileDirectory,
                              informationReport, unsolicitedStatus,
                              eventNotification]
```

13. Access Control and Security Considerations

13.1 SCL-Based Access Control

The System Configuration Language (SCL) defines access control at the **AccessPoint** level in the **IED** element of the **SCD** file:

```
<AccessPoint name="S1">
  <Server>
    <Authentication none="true" password="false" certificate="false"/>
    <LDevice inst="CTRL">
      ...
    </LDevice>
  </Server>
</AccessPoint>
```

13.2 IEC 62351-8 Role-Based Access Control

IEC 62351-8 defines roles for access control in IEC 61850 systems:

Role	Read ST/MX	Write SP/CF	Write SV	Control (CO)	Write SE	Configure RCB
Viewer						
Operator						
Engineer						
Installer						
SecurityAdmin						

13.3 Transport Security

Layer	Mechanism	Standard
TCP	TLS 1.2/1.3	IEC 62351-3
MMS	Authentication via TLS certificates	IEC 62351-4
Application	Role-based access per MMS service	IEC 62351-8

14. Engineering Reference: Address Construction Examples

14.1 Substation One-Line to IEC 61850 Address Mapping

Scenario: Bay1 — 110kV Circuit Breaker CB1, with overcurrent protection, connected to busbar measurement.

IED:	PROT_BAY1	(Protection IED for Bay 1)
LDs:	PROT_BAY1PROT	(Protection functions)
	PROT_BAY1CTRL	(Control and switching)
	PROT_BAY1MEAS	(Measurement)

Physical Item	Function	ACSI Address	FC	R/W
CB1 Status	Position (open/close)	PROT_BAY1CTRL/XCBR1.Pos.stVal	ST	R
CB1 Close	Control action	PROT_BAY1CTRL/XCBR1.Pos.Oper	CO	W (Control)
CB1 Open	Control action	PROT_BAY1CTRL/XCBR1.Pos.Oper	CO	W (Control)
Phase A Current	Measurement	PROT_BAY1MEAS/MMXU1.A.phsA.cVal.mag.f	MX	R
OC Pickup	Protection setpoint	PROT_BAY1PROT/PTOC1.StrVal.setMag.f	SP	R/W
OC Time Dial	Protection setpoint	PROT_BAY1PROT/PTOC1.TmMult.setMag.f	SP	R/W
OC Start	Protection status	PROT_BAY1PROT/PTOC1.Str.general	ST	R
OC Trip	Protection output	PROT_BAY1PROT/PTOC1.Op.general	ST	R
Bus Voltage	Measurement	PROT_BAY1MEAS/MMXU1.PhV.phsA.cVal.mag.f	MX	R

14.2 MMS Wire Format Reference for Above Addresses

ACSI Address	MMS Domain	MMS Item ID
PROT_BAY1CTRL/XCBR1.Pos.stVal [ST]	PROT_BAY1CTRL	XCBR1\$ST\$Pos\$stVal
PROT_BAY1CTRL/XCBR1.Pos.Oper.ctlVal [CO]	PROT_BAY1CTRL	XCBR1\$CO\$Pos\$Oper\$ctlVal

ACSI Address	MMS Domain	MMS Item ID
PROT_BAY1MEAS/MMXU1.A.phsA.cVal.mag.f [MX]	PROT_BAY1MEAS	MMXU1\$MX\$A\$phsA\$cVal\$mag\$f
PROT_BAY1PROT/PTOC1.StrVal.setMag.f [SP]	PROT_BAY1PROT	PTOC1\$SP\$StrVal\$setMag\$f
PROT_BAY1PROT/PTOC1.Str.general [ST]	PROT_BAY1PROT	PTOC1\$ST\$Str\$general

14.3 Complete Read/Write Operation Examples

Read Phase A Current (MX — polling):

```
MMS ReadRequest:
  domainId: "PROT_BAY1MEAS"
  itemId:   "MMXU1$MX$A$phsA$cVal$mag$f"

MMS ReadResponse:
  accessResult: data = FLOAT32(245.7)
```

Read with Quality:

```
MMS ReadRequest:
  domainId: "PROT_BAY1MEAS"
  itemId:   "MMXU1$MX$A$phsA"    (read entire phsA structure: cVal + q + t)

MMS ReadResponse:
  accessResult: data = STRUCTURE {
    cVal: STRUCTURE { mag: STRUCTURE { f: FLOAT32(245.7) }, ang: STRUCTURE { f: FLOAT32(15.3) } },
    q:    BIT-STRING(0x0000),    -- validity=good, source=process
    t:    UTCTIME(...)
  }
```

Write Protection Setpoint (SP — direct):

```
MMS WriteRequest:
  domainId: "PROT_BAY1PROT"
  itemId:   "PTOC1$SP$StrVal$setMag$f"
  data:     FLOAT32(1.25)

MMS WriteResponse:
  success (empty)
```

Control — Close CB1 (Direct Operate):

```
MMS WriteRequest:
  domainId: "PROT_BAY1CTRL"
  itemId:   "XCBR1$CO$Pos$Oper"
  data: STRUCTURE {
    ctlVal: BOOLEAN(TRUE),          -- TRUE = close
    origin: STRUCTURE {
      orCat:  INTEGER(3),           -- remote
      orIdent: OCTET-STRING("SCADA_01")
    },
    ctlNum:  INTEGER(42),
    T:       UTCTIME(now),
```

```
Test:    BOOLEAN(FALSE),
Check:   BIT-STRING(0b11)    -- interlockCheck + synchroCheck
}

MMS WriteResponse:
success
```

15. Architectural Rules and Best Practices

15.1 The Three Golden Rules

Rule 1 — FC determines access, not the DO type. A Data Object has no intrinsic read/write designation. Only its Data Attributes, partitioned by FC, have access rules. The same DO (**Pos**) contains read-only (**ST**), read-write (**CO**), and configuration (**CF**) DAs.

Rule 2 — Never write process values directly. **FC=ST** and **FC=MX** DAs represent the live truth of the power system as measured by transducers and IED firmware. They are never client-writable. For testing, use **FC=SV** substitution with **subEna=TRUE**. For control, use the ACSI control model.

Rule 3 — Control is a state machine, not a write. Any integration that issues **SetDataValues** directly to control output DAs will be rejected by any conformant IEC 61850 server. The ACSI control model enforces interlocking, authorization, timeout, and audit trail at the protocol level.

15.2 Performance Considerations

Pattern	Use Case	Recommendation
GetDataValues polling	Engineering tools, one-time reads	Acceptable, max 1 req/s per IED
GetDataSetValues	SCADA periodic scan	Preferred over individual reads
RCB subscription	Real-time monitoring (ST, MX)	Mandatory for operational SCADA
GOOSE	Protection logic, fast tripping	Use for <4ms response; not MMS
Sampled Values (SV)	Merging unit measurements	IEC 61850-9-2, not MMS

15.3 Common Integration Errors

Error	Cause	Correct Approach
Direct write to \$ST\$Pos\$stVal	Misunderstanding FC=ST This is read-only; use control model	Direct write to \$CO\$Pos\$Oper\$ctlVal (SetDataValues) Bypassing control model Write to \$CO\$Pos\$Oper as STRUCTURE, not leaf ctlVal
Missing \$ in MMS item ID	Wrong separator ACSI uses ., MMS uses \$	Reading wrong FC Querying \$MX instead of \$ST for status Check CDC definition for correct FC
SE write without SGCB EDIT	Writing SE DA without SelectEditSG	Always call SelectEditSG first
No RCB reservation	Multiple clients competing for RCB	Write Resv=TRUE before configuring RCB
Control without synchrocheck	Setting Check bitmask to 0	Set appropriate Check bits per station design

16. Quick Reference Summary Tables

16.1 FC Quick Reference

FC	Write via SetDataValues	Write via Control	Notes
ST	✗	✗	Process read-only
MX	✗	✗	Measurement read-only
SP	✓	✗	Direct write
SV	✓	✗	Enable via subEna
CF	✓	✗	Engineering access
DC	✗	✗	Static
EX	✗	✗	Vendor static
SE	✓	✗	Requires SGCB EDIT context
CO	✗	✓	ACSI control model only
OR	✗	✗	Internal audit
BL	⚠	✗	Implementation-defined
BR	✓	✗	RCB configuration
RP	✓	✗	RCB configuration

16.2 Address Separator Quick Reference

| Context | Separator | Example | | ----- | ----- | ----- |
--- | | ACSI object reference | / between LD and LN; . between DO/DA | IED1CTRL/XCBR1.Pos.stVal | |
MMS item identifier | \$ between all segments | XCBR1\$ST\$Pos\$stVal | | SCL/XML reference | / then \$ notation | IED1CTRL/XCBR1\$ST\$Pos\$stVal | | Dataset member | Full ACSI reference | IED1CTRL/XCBR1.Pos.stVal[ST] |

16.3 Minimum Read/Write Client Checklist

Before reading:

- ☐ Correct MMS domain name (= IED name + LD instance)
- ☐ Correct LN class and instance
- ☐ Correct FC for the data you want (ST for status, MX for measurements)
- ☐ Correct DA path using \$ separators in MMS
- ☐ MMS connection established with correct parameters

Before writing SP/CF/SV:

- ☐ FC is writable (SP, SV, CF, SE)
- ☐ Client has write role (IEC 62351-8)

- ☐ LN `Mod.stVal` is ON or TEST
- ☐ Value type matches DA type (FLOAT32, BOOLEAN, INT32, etc.)
- ☐ Value within defined range (check `$CF$xxx$range`)

Before control (CO):

- ☐ Read `ctlModel` from `$CF$Pos$ctlModel` to determine model (1-4)
- ☐ For SBO: send `SBO` write first, wait for success
- ☐ Populate full Oper structure (ctlVal, origin, ctlNum, T, Test, Check)
- ☐ Handle AddCause in response
- ☐ Monitor ST status change via RCB after operate

Before SE write:

- ☐ Read `LLN0$SG$SGCB$NumOfSG` to confirm group count
- ☐ Write `LLN0$SG$SGCB$EditSG` = target group number
- ☐ Perform all SE DA writes
- ☐ Write `LLN0$SG$SGCB$CnfEdit` = TRUE to commit

End of Document

Document: IEC 61850 MMS Protocol — Rules for Reading and Writing to IEC Addresses **Standard:** IEC 61850 Ed.2 / Ed.2.1 **Classification:** Technical Reference